

Ray Augmented Supervision for 3D Multi-view Object Detection

Huy-Hoang Duong^{1,2} · Adrian Voicila¹ · Guillaume Allibert²

(1) AMPERE SOFTWARE TECHNOLOGY (RENAULT GROUP) (2) UNIVERSITÉ CÔTE D'AZUR · CNRS · I3S



GROUPE
RENAULT

cnrs

UNIVERSITÉ
CÔTE D'AZUR

i3s
sophia antipolis

MOTIVATION

O2M transformed 2D DETR training. Why is 3D still mostly one-to-one?

The missing bridge is an assignment rule that tolerates depth ambiguity and improves the recurrent query stream used at inference.

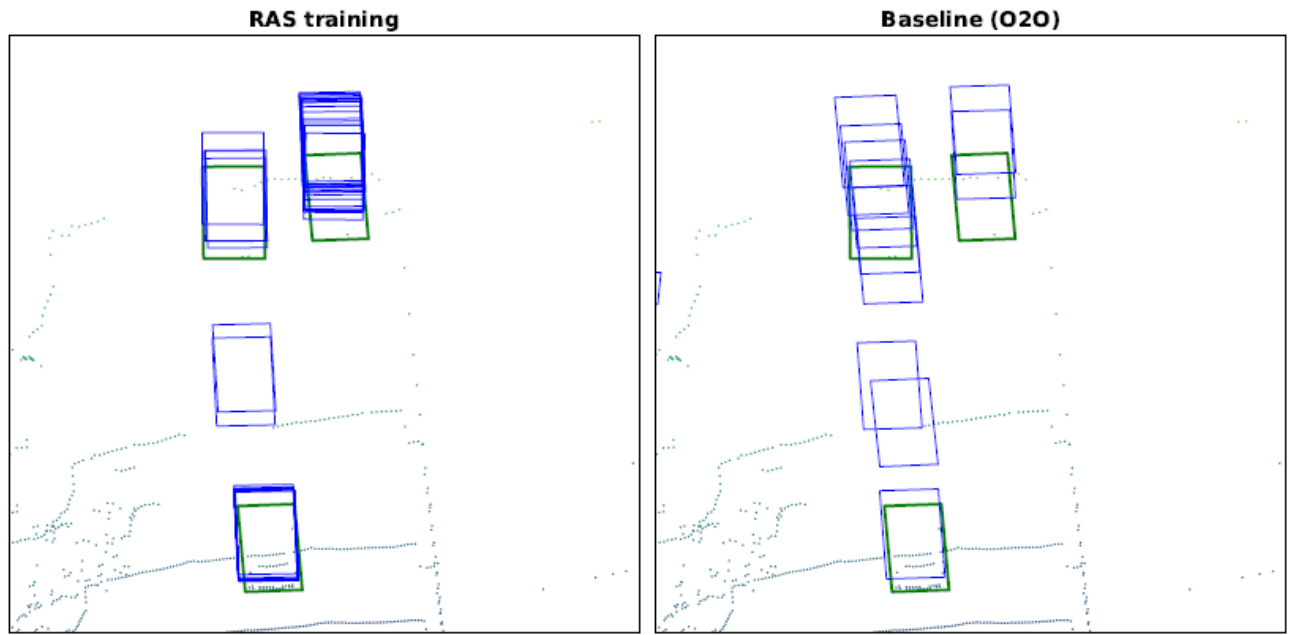
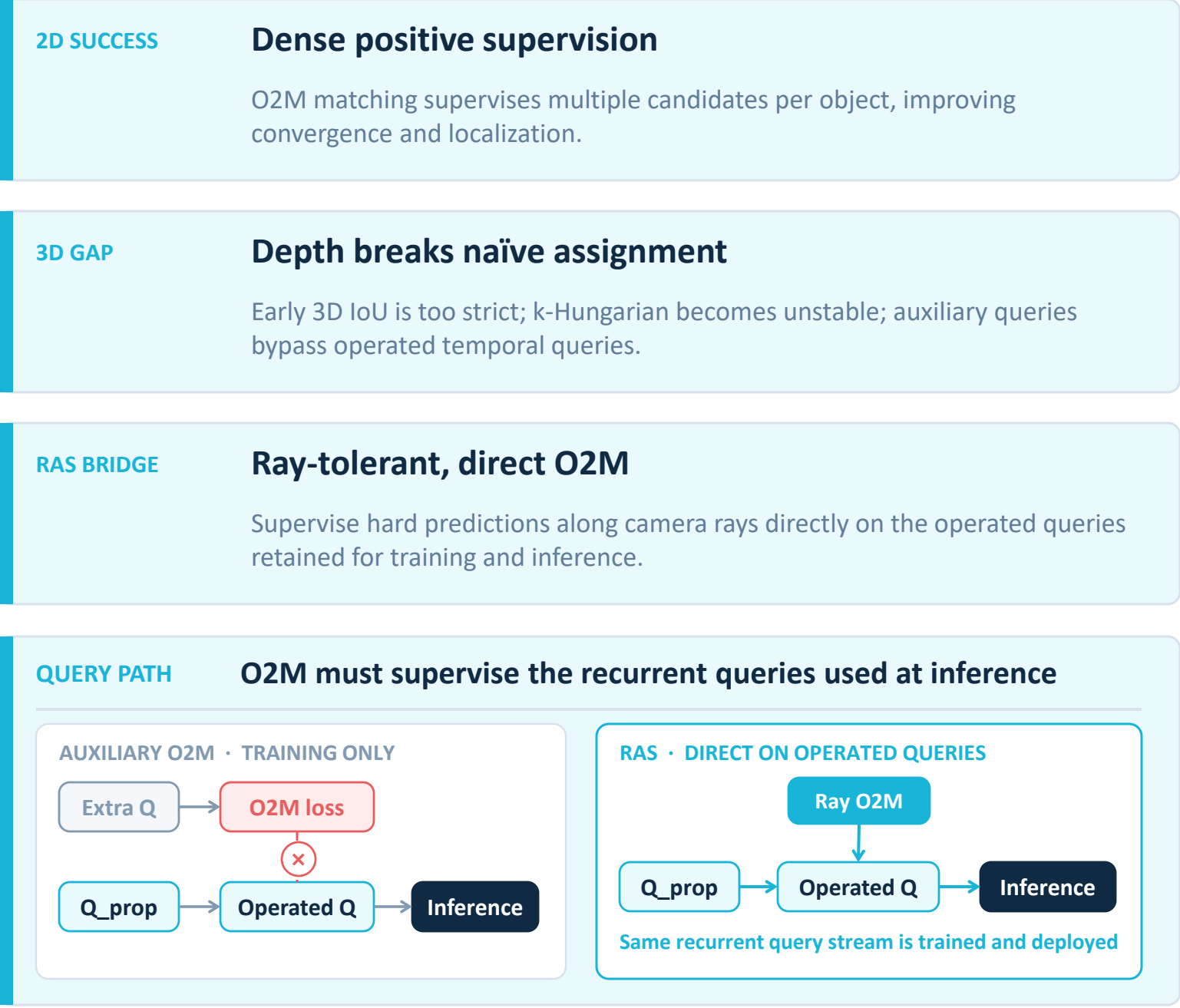


Fig 1 · RAS pushes hard false positives closer to ground truth than O2O, allowing self-attention to eliminate them more effectively

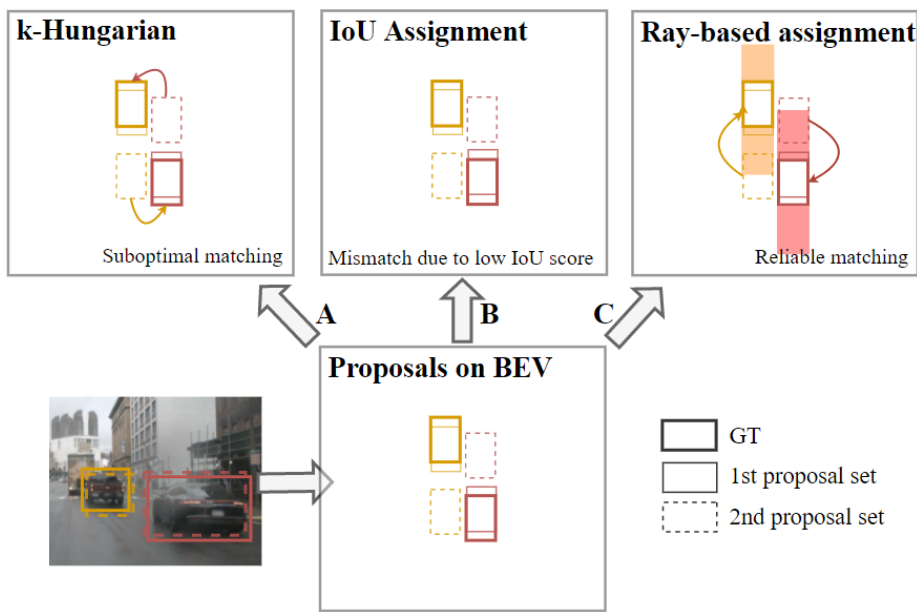
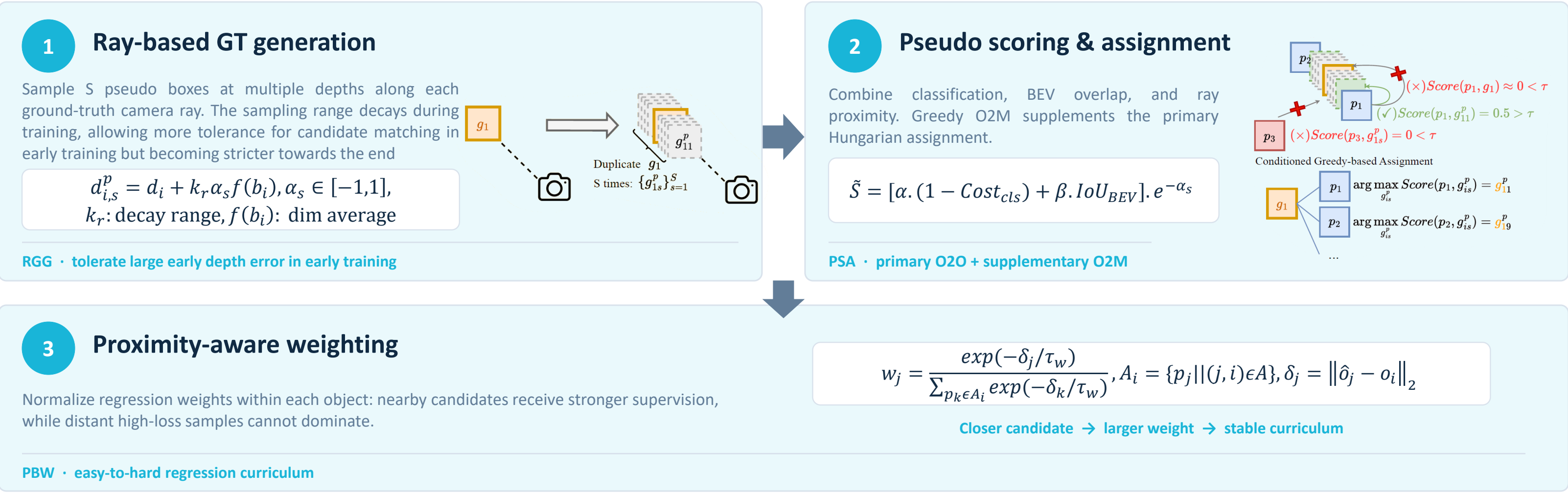


Fig 2 · k-Hungarian or IoU-based assignment can yield suboptimal matching. RAS assigns positive candidates along the camera ray.

METHOD

Three components make O2M reliable in 3D

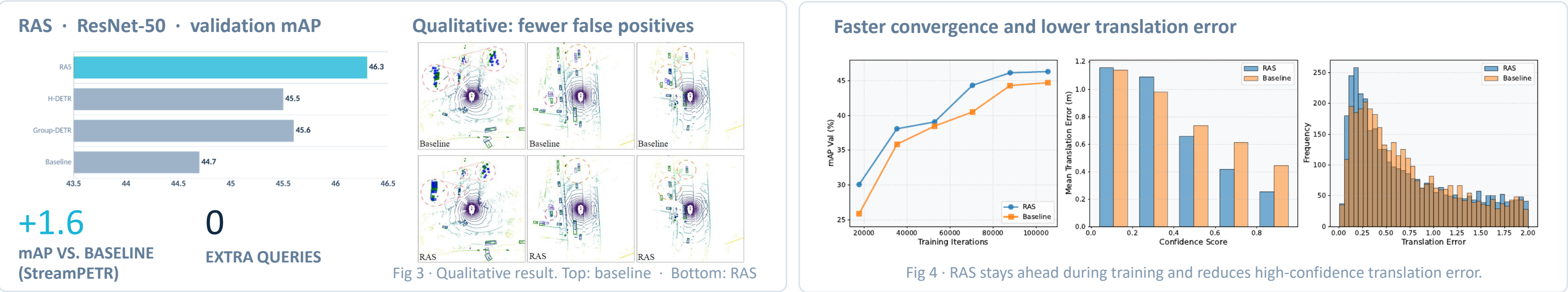
TL,DR: A **plug-and-play one-to-many training scheme** applied directly to operated queries for **Recurrent Transformer in Multiview 3D Detection**.



Cross-attention proposes candidates → self-attention + O2O supervision suppresses duplicates

RESULTS ON NUSCENES

Direct ray supervision improves accuracy without extra queries



GENERALIZES	50.3 (+1.1) mAP · VoVNet-99	45.7 (+0.9) mAP · SparseBEV	47.9 (+1.0) mAP · RayDN + depth	+3.2 ms PER FRAME (NMS inclusion)	46.3 mAP · 55.2 NDS ResNet-50
-------------	--------------------------------	--------------------------------	------------------------------------	--------------------------------------	----------------------------------

TAKEAWAY & LIMITATION

RAS bridges 2D O2M success to 3D by making assignment ray-aware and query-direct. The current end-to-end detector still requires NMS at inference; NMS-free detection with O2M remains to be studied.

huy-hoang.duong@ampere.cars